Harsh Ranjan

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EDUCATION

University of California, Santa Cruz 2022 – Dec 2026
B.S. Computer Engineering, Computer Systems Track *Santa Cruz, CA*

- **Coursework:** Data Structures & Algorithms, Operating Systems, Computer Architecture, Embedded Systems, ML

University of California, San Diego – Summer Session | *Computer Architecture with Lab* Summer 2025

EXPERIENCE

Renewable Resilient Energy Systems Lab, UC Santa Cruz Sep 2026 – Present
Research Intern – Electrical Anomaly Detection System (EADS) *Santa Cruz, CA*

- Improving an **electrical anomaly detection system** under **Dr. Pat Mantey** on distributed **AnyLog / EdgeLake** nodes.
- Work with cross-functional **software and infrastructure** teams, debugging the **AnyLog 2.0** master node deployment.

UC Santa Cruz – Economics & Applied Mathematics Fall 2025 – Present
Tutor and Reader *Santa Cruz, CA*

- Lead tutoring for groups of **10** in **AM 11A** and grade coursework for **AM 10**, **AM 20**, and **STATS 5**.

Tesphase Apr 2025 – Sep 2025
Chief Technology Officer *San Francisco, CA*

- **Winner, SC Launchpad 2025.** Owned the roadmap for a renewable-energy **React Native / Node.js / Supabase** app.
- Ran **30+** NSF I-Corps customer interviews, then shipped **authentication**, **real-time sync**, and **push notifications**.

Balat Enterprises Jun 2024 – Sep 2024
Product Management Intern *Chennai, India*

- Turned customer feedback into **product specs** for a **React / Firebase** app, cutting post-release bugs by **40%.**
- Built an internal **Python (.py)** and **JavaScript (.js)** tool to collect and organize potential customer leads.
- Redesigned an internal **customer-service tool**, cutting response times and lifting user retention by **30%.**

Vyavshay Technologies Jul 2022 – Aug 2022
Marketing Intern *Chennai, India*

- Led **market research** and **campaign analysis** for a rural **cab management app**, lifting user retention by **15%.**

PROJECTS

Microgrid Manager – Edge AI Energy Disaggregation **Python, PyTorch, TensorFlow, ONNX, Kafka, Docker**

- Architected an edge-only AI **NILM** system for a live solar microgrid, disaggregating **14** appliance loads across **3** panels.
- Streamed telemetry over **Kafka** and **MQTT** (**~960 msg/min**) into **3 EdgeLake / AnyLog** nodes and **PostgreSQL**.
- Replaced failing LLM labels (**100%** false negatives) with **physics-informed labels** and trained **3 BiLSTMs** on **705K+** rows.
- Served **ONNX** models at **single-digit ms** latency and wrote a blog on the system for the **EdgeLake Foundation**.

GPT-2 CPU Performance Optimization **C, Linux perf, AVX2, OpenMP**

- Cut the L1 data-cache miss rate in GPT-2 training kernels from **~9% to 2–3%** with **cache tiling** and **AVX2 SIMD**.

ESP32 Weather Station **Rust, ESP32-C3, Raspberry Pi, HTTP**

- Programming an **ESP32-C3** in **Rust** to post **wttr.in** and onboard **sensor** readings to a **Raspberry Pi 4** HTTP server.

HTTP/HTTPS Proxy Server **C, POSIX threads, Verilog**

- Engineered a concurrent **C** proxy with **CONNECT** tunneling and a **FIFO / LRU** cache, cutting cached response time **50%.**
- Porting the server to **FPGA** hardware in **Verilog** on **Spartan-3E**, **Basys 3**, and **Alchitry Io** boards.

Osmosis – VGA Molecule Game **FPGA, Verilog, Vivado**

- Designed a **640x480 VGA** game in **Verilog** across **19** modules with **LFSR** motion, **FSM** states, and **zero** sync errors.

TECHNICAL SKILLS

Languages: C, C++, Python, Rust, JavaScript (.js), SQL, Verilog, SystemVerilog, Assembly, Shell / Bash
ML & Data: PyTorch, TensorFlow, ONNX, BiLSTM, snnTorch, Kafka, MQTT, PostgreSQL, AnyLog / EdgeLake
Tools: React, React Native, Node.js, FastAPI, Supabase, Firebase, Docker, Git, CI/CD, Linux perf, OpenMP
Hardware: FPGA, RTL Design, Vivado, Basys 3, ESP32, Raspberry Pi, UART, SPI, I2C, JTAG, GDB
Other: Performance-Based Programming, Market Research, Customer Discovery, Product Specs, Technical Writing